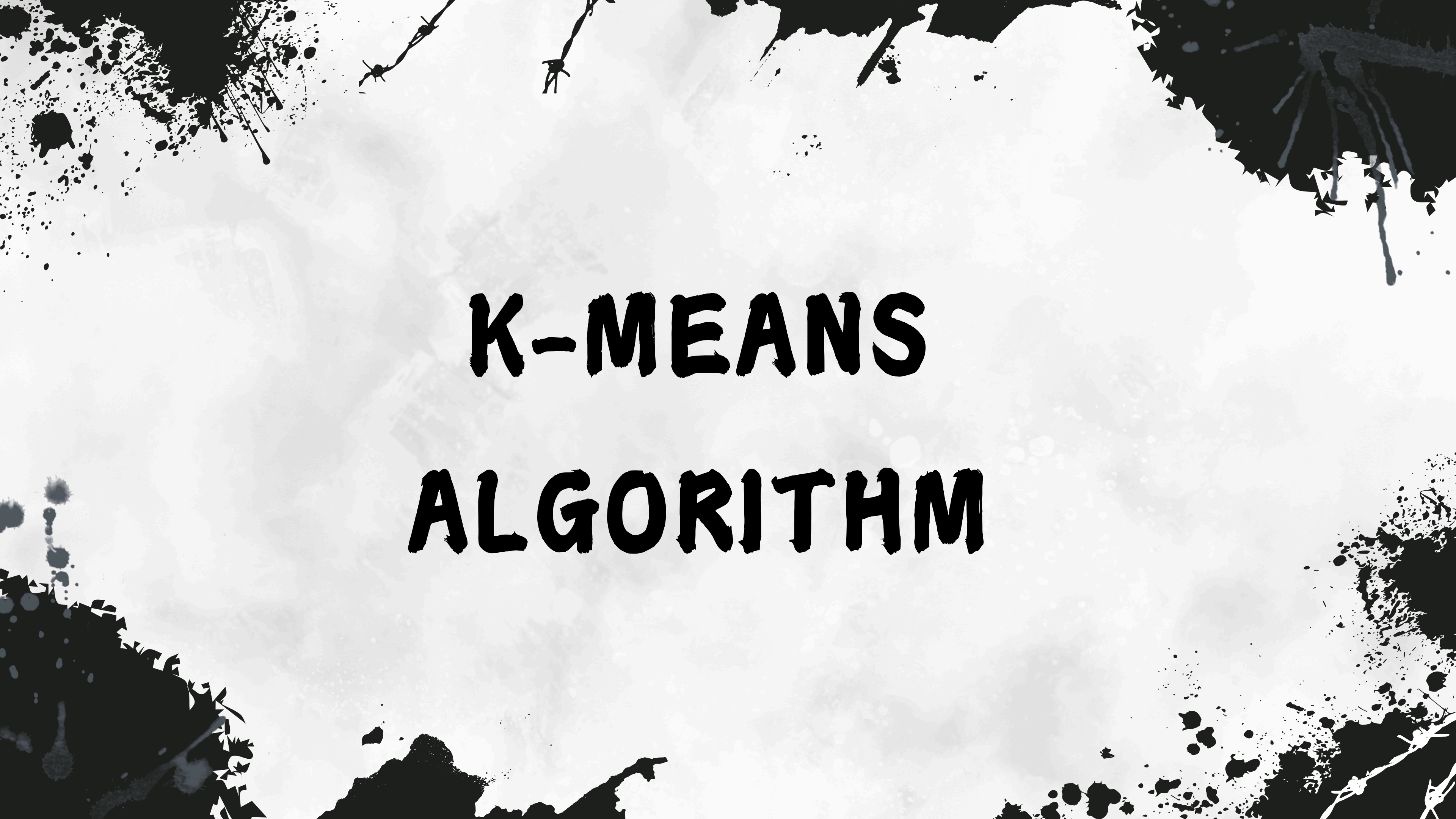




K-MEANS ALGORITHM

Cluster Analysis: Basic Concepts and Methods

- Cluster Analysis: Basic Concepts
 1. Partitioning Methods
 2. Hierarchical Methods
 3. Density-Based Methods
- Summary

What is Cluster Analysis?

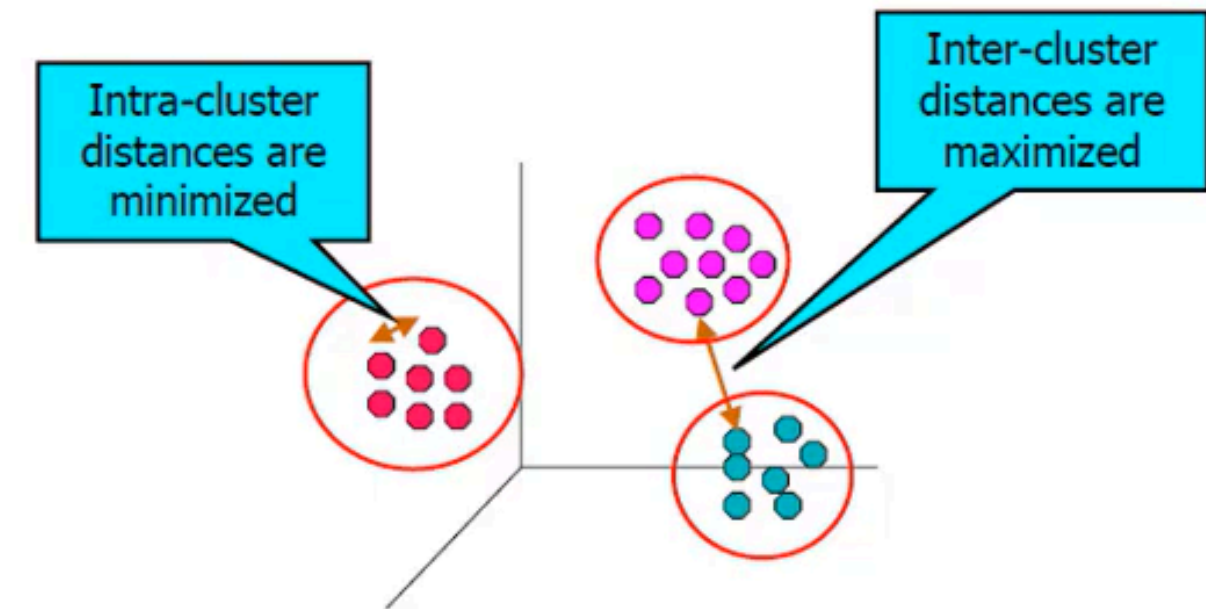
- **Cluster**: A collection of data objects
 - similar (or related) to one another **within the same group**
 - dissimilar (or unrelated) to the objects in other groups
- **Cluster analysis** (or *clustering, data segmentation, ...*)
 - Finding **similarities between data** according to the characteristics found in the data and grouping similar data objects into clusters
- **Unsupervised learning**: no predefined classes (i.e., *learning by observations* vs. learning by examples: supervised)
- **Typical applications**
 - As a **stand-alone tool** to get insight into data distribution
 - As a **preprocessing step** for other algorithms

Clustering as a Preprocessing Tool (Utility)

- Summarization:
 - Preprocessing for regression, PCA, classification, and association analysis
- Compression:
 - Image processing
- Finding K-nearest Neighbors
 - Localizing search to one or a small number of clusters
- Outlier detection
 - Outliers are often viewed as those “far away” from any cluster

Quality: What Is Good Clustering?

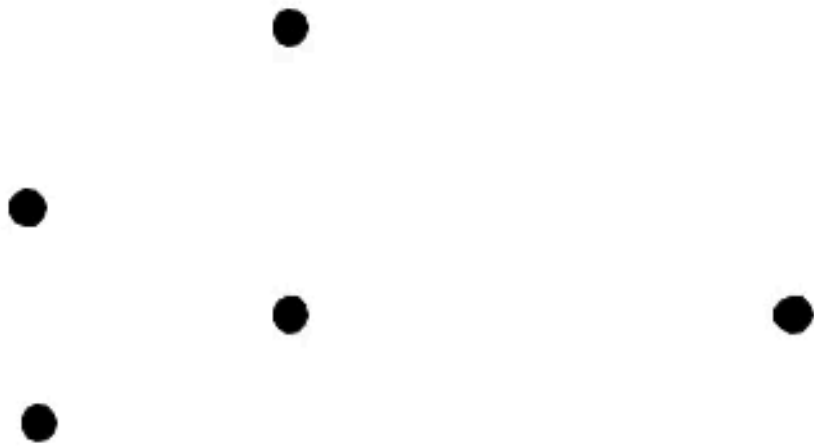
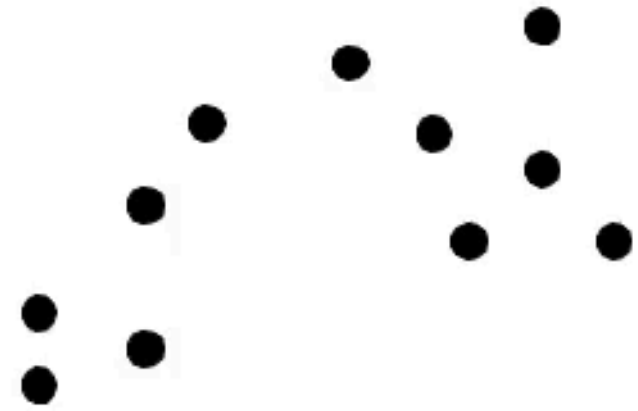
- A **good clustering** method will produce high quality clusters
 - high intra-class similarity: **cohesive** متماسك within clusters
 - low inter-class similarity: **distinctive** متميز between clusters
- The quality of a clustering method depends on:
 - the **similarity measure** used by the method
 - its **implementation**, and
 - Its **ability to discover** some or all of the hidden patterns



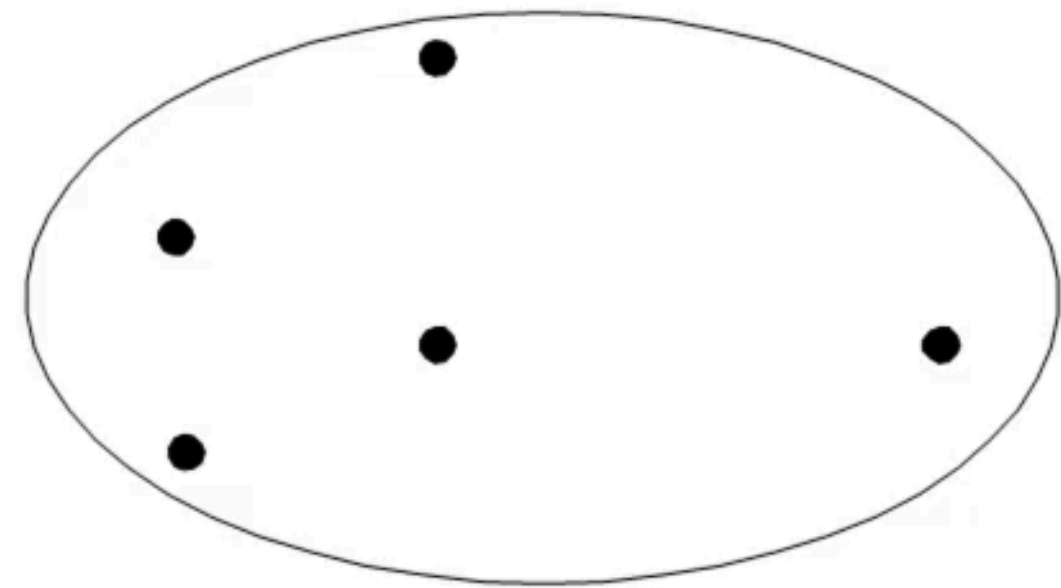
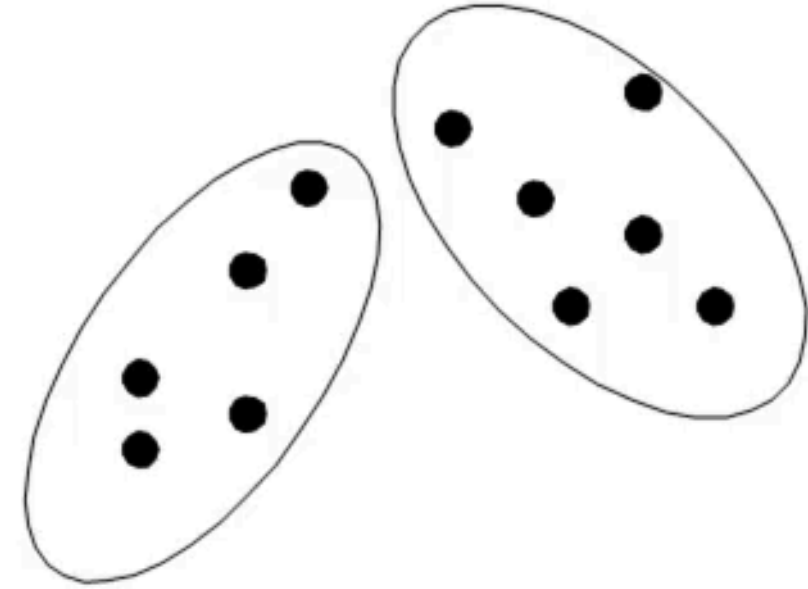
Un-Supervising Learning

- Cluster Analysis: Basic Concepts
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Partitional Clustering



Original Points

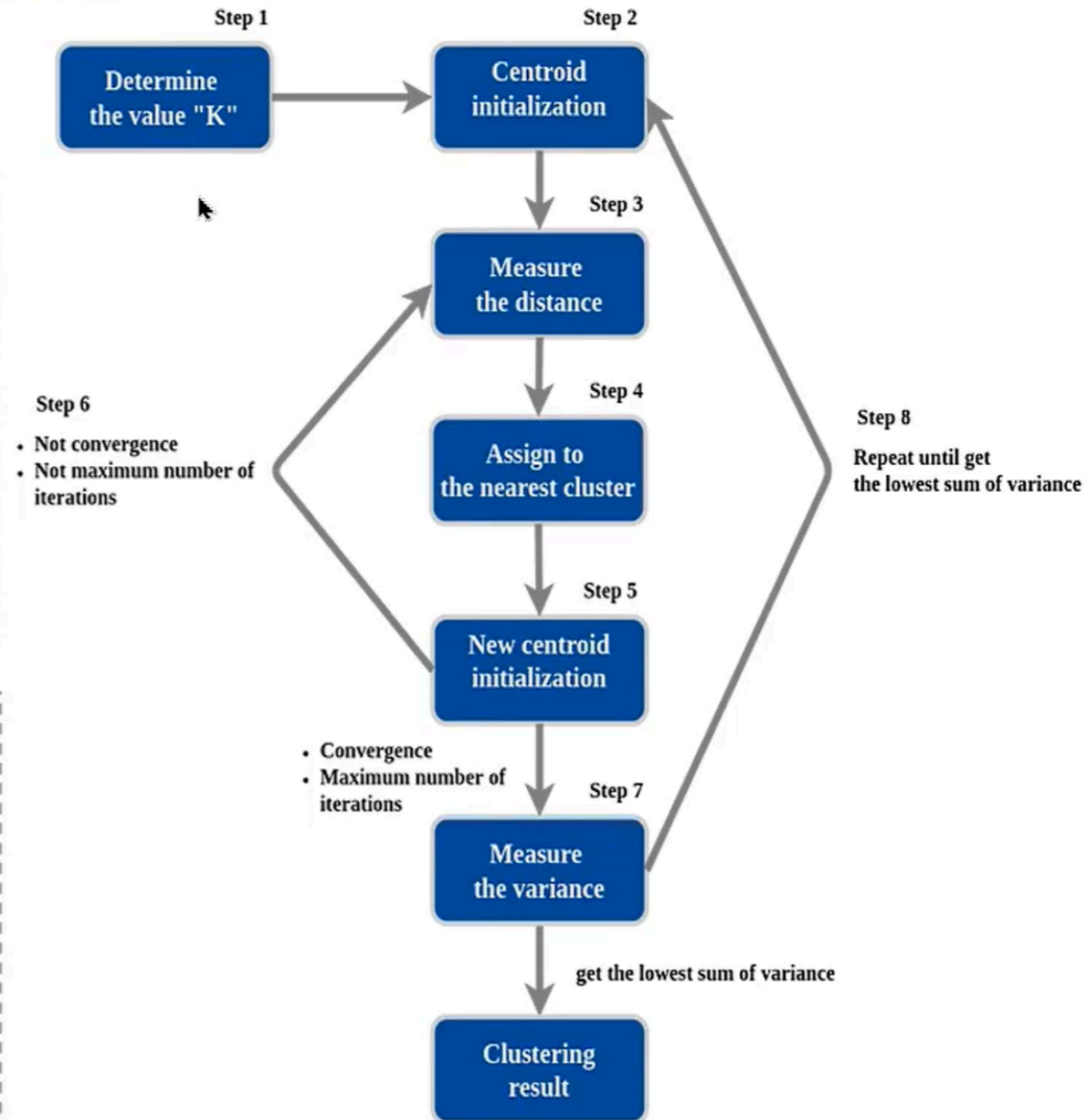
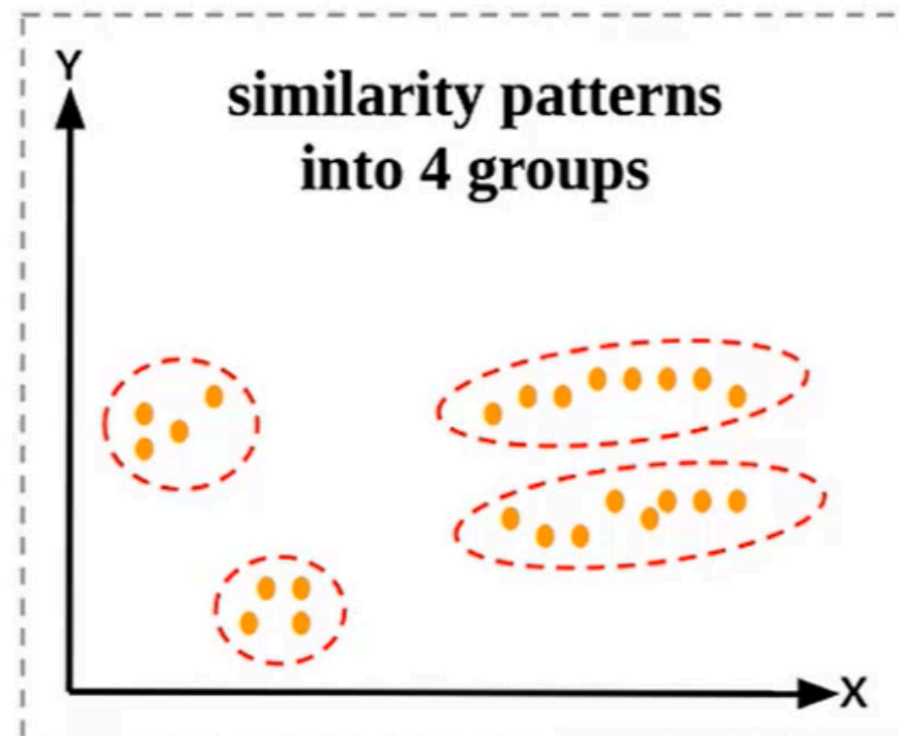
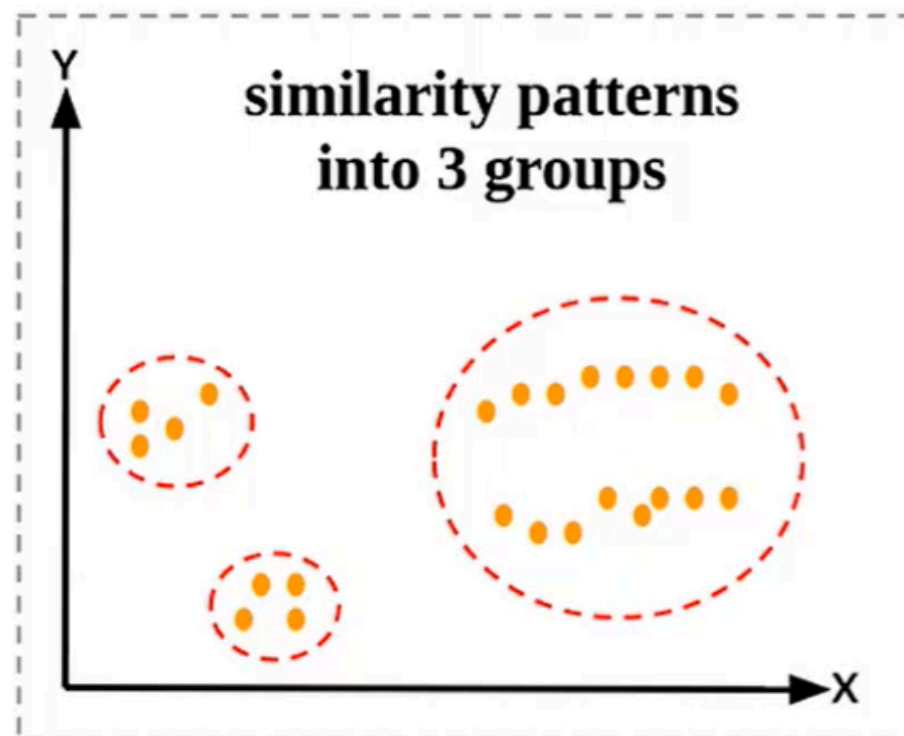
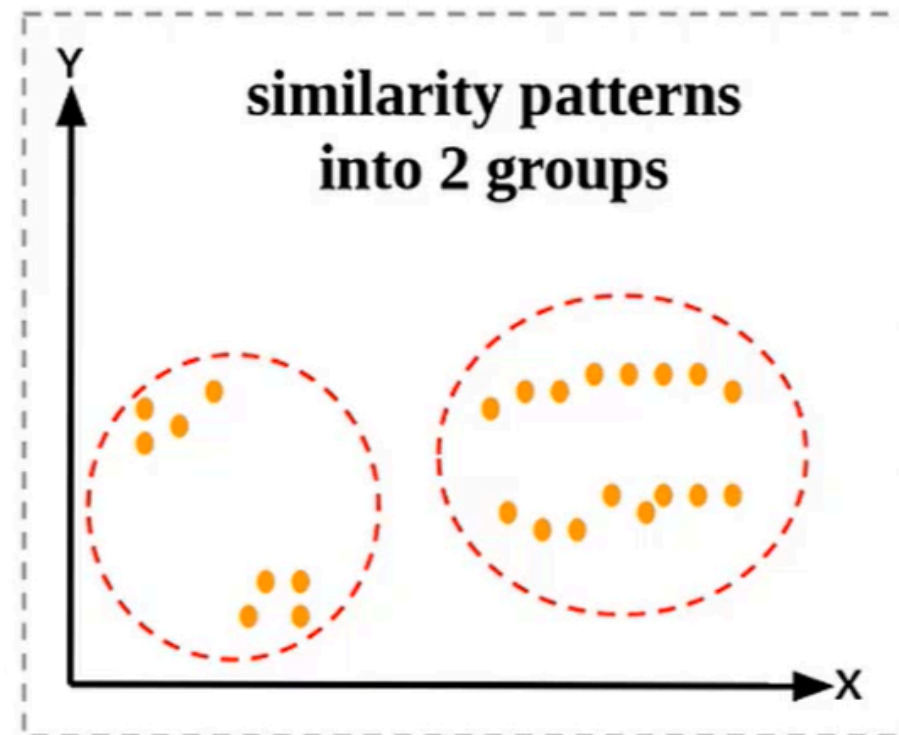
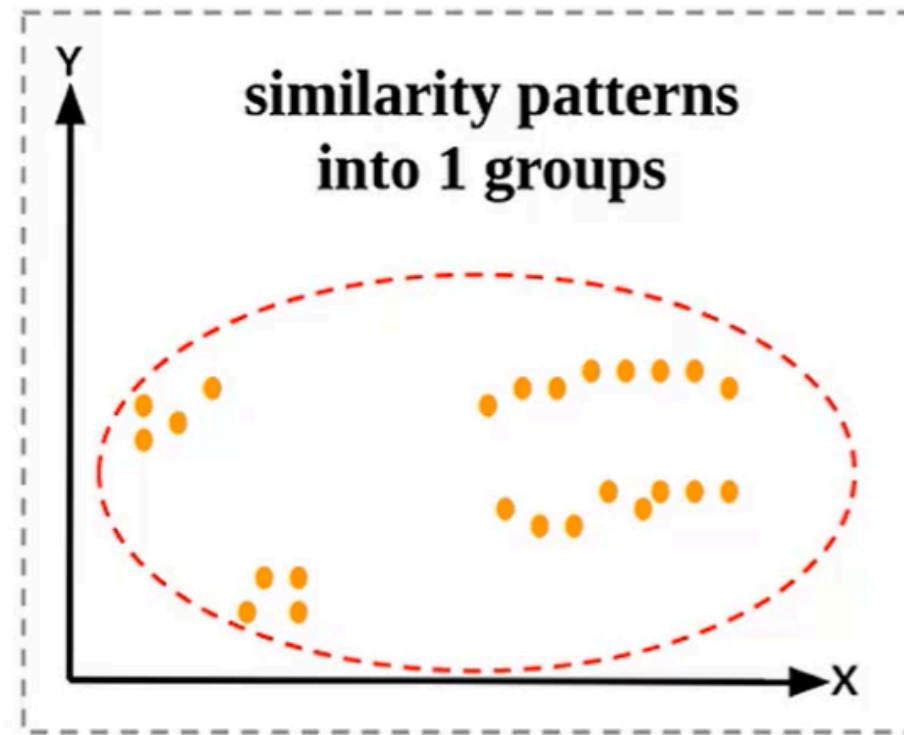


A Partitional Clustering

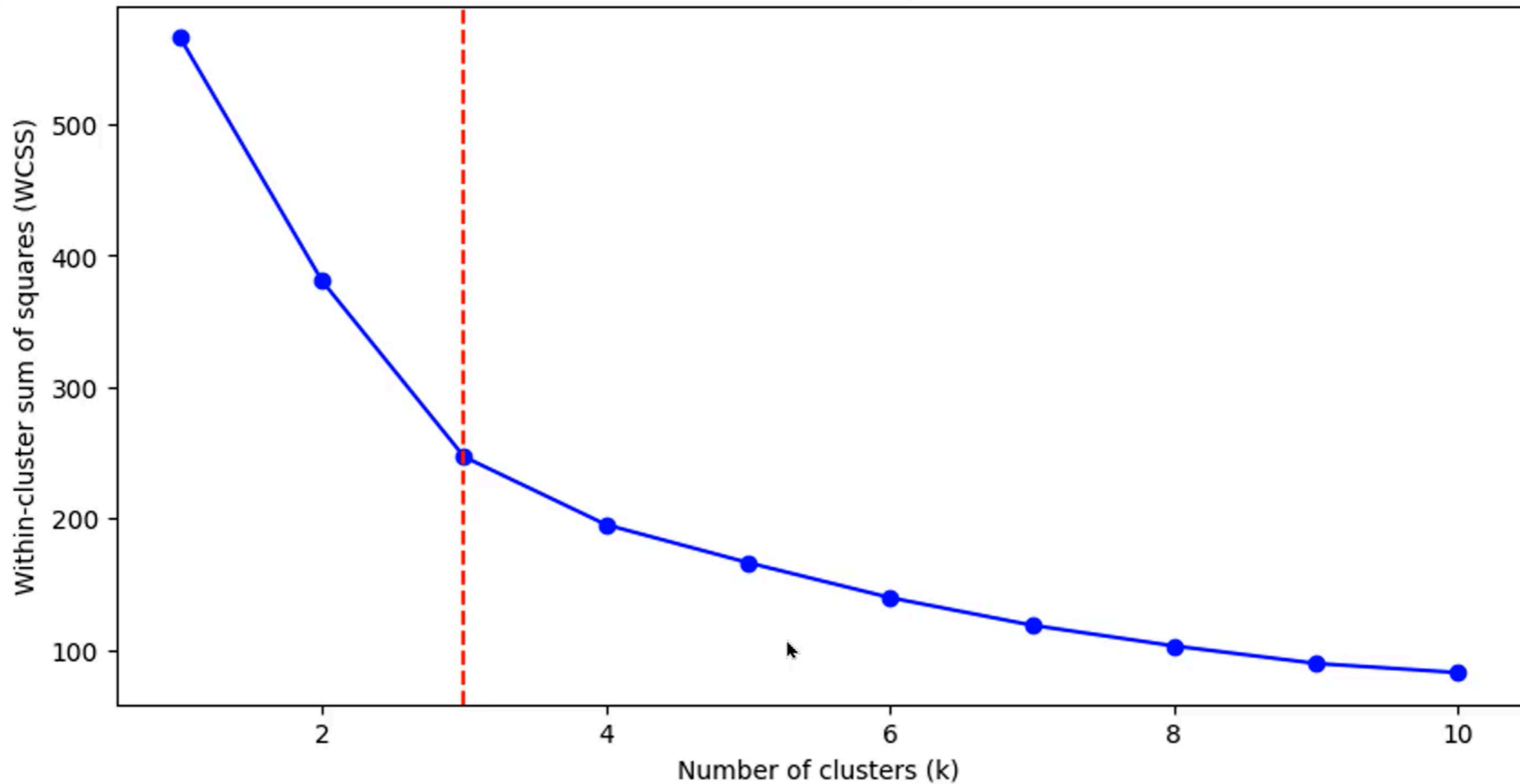
1-The K-Means Clustering Method

- Given k , the *k-means* algorithm is implemented in four steps:
 - **Partition objects** into k nonempty subsets
 - **Compute seed points** as the centroids of the clusters of the current partitioning (the centroid is the center, i.e., *mean point*, of the cluster)
 - **Assign each object** to the cluster with the nearest seed point
 - **Go back to Step 2**, stop when the assignment does not change

1-The K-Means Clustering Method ... cont.

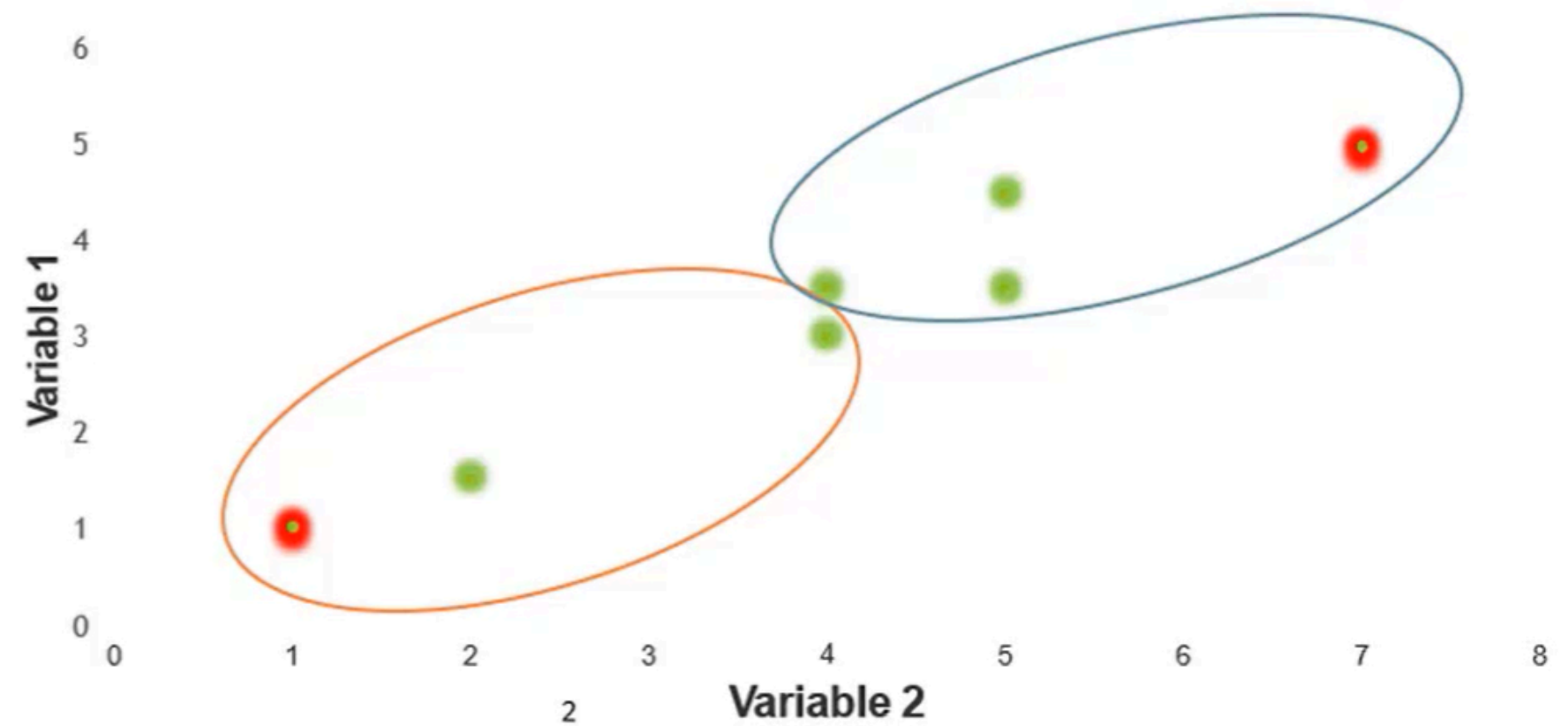


Elbow Method for Optimal k



A Simple example k-means (using $K=2$)

Individual	Variable 1	Variable 2
1	1	1
2	1.5	2
3	3	4
4	5	7
5	3.5	5
6	4.5	5
7	3.5	4.5



Step 1:

- Initialization: Randomly we choose following two centroids ($k=2$) for two clusters. In this case the 2 centroid are: $m1=(1.0,1.0)$ and $m2=(5.0,7.0)$.

	Individual	Mean Vector
Group 1	1	(1.0, 1.0)
Group 2	4	(5.0, 7.0)

Step 2:

	Centroid 1	Centroid 2
1	$\sqrt{(1 - 1)^2 + (1 - 1)^2} = 0$	$\sqrt{(5 - 1)^2 + (7 - 1)^2} = 7.21$
2	$\sqrt{(1 - 1.5)^2 + (1 - 2)^2} = 1.12$	$\sqrt{(5 - 1.5)^2 + (7 - 2)^2} = 6.10$
3	$\sqrt{(1 - 3)^2 + (1 - 4)^2} = 3.61$	$\sqrt{(5 - 3)^2 + (7 - 4)^2} = 3.61$
4	$\sqrt{(1 - 5)^2 + (1 - 7)^2} = 7.21$	$\sqrt{(5 - 5)^2 + (7 - 7)^2} = 0$
5	$\sqrt{(1 - 3.5)^2 + (1 - 5)^2} = 4.72$	$\sqrt{(5 - 3.5)^2 + (7 - 5)^2} = 2.5$
6	$\sqrt{(1 - 4.5)^2 + (1 - 5)^2} = 5.31$	$\sqrt{(5 - 4.5)^2 + (7 - 5)^2} = 2.06$
7	$\sqrt{(1 - 3.5)^2 + (1 - 4.5)^2} = 4.30$	$\sqrt{(5 - 3.5)^2 + (7 - 4.5)^2} = 2.92$

Step 2:

- Thus, we obtain two clusters containing: 
- $\{1, 2, 3\}$ and $\{4, 5, 6, 7\}$.
- Their new centroids are:

$$\text{Group 1} = \left(\frac{1+1.5+3}{3} \right), \left(\frac{1+2+4}{3} \right) = (1.83, 2.33)$$

$$\text{Group 2} = \left(\frac{5+3.5+4.5+3.5}{4} \right), \left(\frac{7+5+5+4.5}{4} \right) = (4.12, 5.38)$$

Step 3:

	Centroid 1	Centroid 2
1	$\sqrt{(1.83 - 1)^2 + (2.33 - 1)^2} = 1.57$	$\sqrt{(4.12 - 1)^2 + (5.38 - 1)^2} = 5.38$
2	$\sqrt{(1.83 - 1.5)^2 + (2.33 - 2)^2} = 0.47$	$\sqrt{(4.12 - 1.5)^2 + (5.38 - 2)^2} = 4.29$
3	$\sqrt{(1.83 - 3)^2 + (2.33 - 4)^2} = 2.04$	$\sqrt{(4.12 - 3)^2 + (5.38 - 4)^2} = 1.78$
4	$\sqrt{(1.83 - 5)^2 + (2.33 - 7)^2} = 5.64$	$\sqrt{(4.12 - 5)^2 + (5.38 - 7)^2} = 1.84$
5	$\sqrt{(1.83 - 3.5)^2 + (2.33 - 5)^2} = 3.15$	$\sqrt{(4.12 - 3.5)^2 + (5.38 - 5)^2} = 0.73$
6	$\sqrt{(1.83 - 4.5)^2 + (2.33 - 5)^2} = 3.78$	$\sqrt{(4.12 - 4.5)^2 + (5.38 - 5)^2} = 0.54$
7	$\sqrt{(1.83 - 3.5)^2 + (2.33 - 4.5)^2} = 2.74$	$\sqrt{(4.12 - 3.5)^2 + (5.38 - 4.5)^2} = 1.08$

Therefore,
the new
clusters are:

$\{1,2\}$ and $\{3,4,5,6,7\}$

$$\text{Group 1} = \left(\frac{1+1.5}{2} \right), \left(\frac{1+2}{2} \right) = (1.25, 1.5)$$

$$\text{Group 2} = \left(\frac{3+5+3.5+4.5+3.5}{5} \right), \left(\frac{4+7+5+5+4.5}{5} \right) = (3.9, 5.1)$$

Step 4:

	Centroid 1	Centroid 2
1	$\sqrt{(1.25 - 1)^2 + (1.5 - 1)^2} = 0.58$	$\sqrt{(3.9 - 1)^2 + (5.1 - 1)^2} = 5.02$
2	$\sqrt{(1.25 - 1.5)^2 + (1.5 - 2)^2} = 0.56$	$\sqrt{(3.9 - 1.5)^2 + (5.1 - 2)^2} = 3.92$
3	$\sqrt{(1.25 - 3)^2 + (1.5 - 4)^2} = 3.05$	$\sqrt{(3.9 - 3)^2 + (5.1 - 4)^2} = 1.42$
4	$\sqrt{(1.25 - 5)^2 + (1.5 - 7)^2} = 6.66$	$\sqrt{(3.9 - 5)^2 + (5.1 - 7)^2} = 2.20$
5	$\sqrt{(1.25 - 3.5)^2 + (1.5 - 5)^2} = 4.16$	$\sqrt{(3.9 - 3.5)^2 + (5.1 - 5)^2} = 0.41$
6	$\sqrt{(1.25 - 4.5)^2 + (1.5 - 5)^2} = 4.78$	$\sqrt{(3.9 - 4.5)^2 + (5.1 - 5)^2} = 0.61$
7	$\sqrt{(1.25 - 3.5)^2 + (1.5 - 4.5)^2} = 3.75$	$\sqrt{(3.9 - 3.5)^2 + (5.1 - 4.5)^2} = 0.72$

Therefore, there is no change in the cluster

- Thus, the algorithm comes to a halt here and final result consist of 2 clusters {1,2} and {3,4,5,6,7}.

The K-Means Advantages

Simplicity: K-means is easy to implement and understand, making it a popular choice for clustering tasks, especially for large datasets.

Efficiency: It is computationally efficient and works well with large datasets.

Scalability: K-means can handle large datasets with ease, making it suitable for clustering in big data applications.

Interpretability **قابلية للتفسير**: The clusters produced by k-means are easy to interpret, as each data point is assigned to the cluster with the nearest mean.

Versatility **تنوع فى الداتا**: K-means can be applied to various types of data, including **numerical** and **categorical** data, making it a versatile clustering algorithm.

The K-Means Disadvantages

Sensitive to Initial Centroids: K-means clustering is sensitive to the initial selection of cluster centroids, which can lead to different results for each run.

Assumes Spherical Clusters مجموعات كروية : K-means assumes that clusters are spherical and have similar sizes, which may not always hold true for real-world datasets with irregularly shaped clusters or clusters of varying sizes.

Requires Predefined Number of Clusters: The number of clusters (k) needs to be specified in advance, which may not always be known and choosing an inappropriate value for k can result in suboptimal clustering.

Outlier Sensitivity: K-means is sensitive to outliers, as they can disproportionately affect the positions of cluster centroids and the overall clustering result.

Non-Robust to Noise: K-means may produce poor results when dealing with noisy data or datasets with overlapping clusters.

The background is a high-contrast, abstract composition. It features a light gray, textured surface that resembles watercolor or ink on paper. Overlaid on this are various black elements: large, irregular ink splatters and blotches, particularly in the corners; and several thin, dark, brush-like strokes that sweep across the upper and lower portions of the frame. The overall effect is one of raw, expressive energy.

THANK YOU